Designing, Building and Verifying an Embedded System

Introduction to ARM Cortex-M Microprocessors — Chapter 14

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Why This Chapter Matters

Earlier chapters used one peripheral at a time. A real system samples on a schedule, answers buttons, drives indicators and reports to a host, all at once, on one processor, with no operating system. The hard part is deciding *before* coding which peripherals the job needs and whether the timing can work. Then you must *prove* the result meets the requirement.

After this chapter, you can:

1. Turn a requirement into a decomposition and a peripheral choice.
2. Choose polling, timed polling or interrupts per activity.
3. Structure a cooperative, non-blocking main loop.
4. Protect data shared between ISR and main.
5. Write a timing budget, then verify it by measurement.
6. Diagnose a multi-peripheral system bottom-up.

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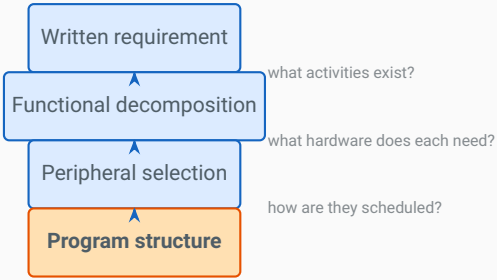
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The Step Before Any Code

A requirement is prose. An architecture is a set of decisions. Getting from one to the other is the engineering, and it happens on paper.



Why this order matters

Choosing peripherals before decomposing the activities means you discover the pin conflict after the board is made.

Four Ways to Structure an Activity

Structure	Use when	Cost
Blocking poll	Never, in a real system	Burns the CPU; blocks everything else
Timed poll	The activity has a known period and loose jitter	Simple; jitter is the loop time
Interrupt	The event is external and the deadline is tight	Needs care with shared data
DMA	Bulk data, no per-byte decision	Setup cost; fewest CPU cycles

The decision is per activity, not per program

One system routinely uses all four: DMA for an ADC stream, an interrupt for a button, a timed poll for a display refresh, and the main loop for everything with no deadline at all.

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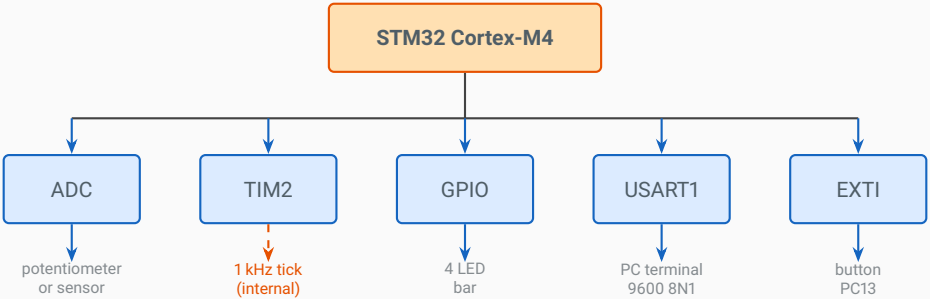
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The Project



Five peripherals, one processor, no OS.

1. Sample the ADC at exactly 10 Hz.
2. Display the level as a 4-LED bar.
3. Transmit each reading over UART as text.
4. Button press toggles report between raw counts and millivolts.
5. Never miss a sample. Never block.

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The Pattern

```
while (1) {
    uint32_t now = millis();

    if (now - last_sensor >= 2000) {    // every 2 s
        last_sensor = now;
        read_sensor();
    }
    if (now - last_display >= 100) {    // every 100 ms
        last_display = now;
        update_display();
    }
    if (button_flag) {                  // set by an ISR
        button_flag = 0;
        handle_button();
    }
}
```

Nothing waits

Each activity asks “is it my turn yet?” and returns immediately if not. The loop runs thousands of times per second, and every activity gets serviced close enough to its deadline.

Why `now - last >= period` and Not `now >= next`

The subtraction form

```
if (now - last >= 2000)
```

Survives the 32-bit counter wrapping after 49 days, because unsigned subtraction wraps consistently.

The comparison form

```
if (now >= next)
```

Breaks at the wrap: `next` is a huge number, `now` has reset to a small one, and the activity stops for 49 days.

A bug you cannot find by testing

It appears once, after seven weeks of continuous running. This is exactly the class of fault that reasoning catches and testing does not — which is why the argument matters more than the code.

Sharing Data With an ISR

Two hazards

- Missing `volatile` — the optimiser caches a value the ISR changes
- Torn access — `main()` reads a multi-word variable while the ISR is halfway through writing it

Three defences

- Mark it `volatile`
- Keep it to a single `uint32_t` — read atomically on a 32-bit machine
- Or disable interrupts briefly around the access

The flag pattern

The ISR does the minimum — clear the hardware flag, set a software one, return. `main()` does the real work. That keeps the handler short and keeps the shared state to one byte.

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The Timing Budget

Write it before implementation, and it tells you whether the design is possible at all.

Activity	Period	Duration	Load
SysTick ISR	1 ms	2 µs	0.2 %
Sensor read	2 s	400 µs	0.02 %
Display update	100 ms	3 ms	3 %
Button ISR	rare	5 µs	≈ 0
Total			≈ 3.3 %

What the number tells you

3 % means comfortable headroom. 80 % means the design is fragile — one slow path and a deadline is missed. Over 100 % means it cannot work, and you know that before writing a line.

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Predict, Then Measure

The discipline

1. Predict the number — period, duty cycle, execution time
2. Measure it
3. When they disagree, find out why

How to measure

- Set a GPIO pin high on entry, low on exit — watch it on a scope
- Read `SysTick->VAL` before and after
- Toggle a pin per loop iteration and measure the frequency

The disagreement is the finding

If you predicted 400 µs and measure 3 ms, you have learned something — about the compiler, the clock tree, or your own assumptions.

Measuring without predicting teaches you nothing, because any number looks plausible.

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What a Working Demonstration Does Not Prove

It runs. What is still unknown?

- Whether it survives the counter wrap at 49 days
- Whether the deadline is met in the worst case, not the typical one
- Whether a shared variable is ever torn
- What happens when a sensor fails rather than reads wrong
- Whether it works at the temperature extremes

Demonstration is evidence, not proof

A demo shows the system works once, here, now, under the conditions you happened to create.

Everything else you believe about it comes from argument: the timing budget, the wrap analysis, the reasoning about shared state.

A Debugging Methodology

When a multi-peripheral system misbehaves, work bottom-up. Each step either finds the fault or eliminates a layer.

Step	Question
1	Is the clock enabled for this peripheral?
2	Is the pin in the right mode, with the right alternate function?
3	Does the register read back what you wrote?
4	Does the peripheral's status flag ever set?
5	Does the ISR run — toggle a pin inside it and look
6	Is the shared variable `volatile`, and read atomically?
7	Does the timing match the budget you predicted?

Steps 1 and 2 account for most first-time faults — and a register that does not read back is an unclocked peripheral.

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What You Have Learned

The mechanism

- A CPU fetches, decodes, executes — forever
- Memory is numbered; some numbers are hardware
- C reaches them through pointers
- Instructions are what your C becomes

The practice

- Enable the clock first
- `volatile` on everything hardware touches
- Read–modify–write preserves the neighbours
- Interrupts for deadlines, timers for time
- Predict, then measure

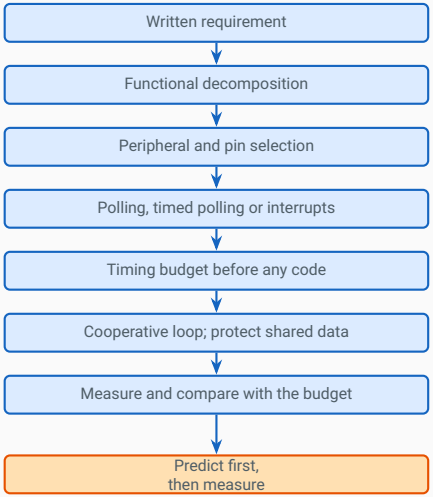
The one sentence from Chapter 1

The CPU has no special instruction for “turn on an LED”. It has instructions that read and write addresses — and some of those addresses are wired to hardware.

Everything in the thirteen chapters since has been an elaboration of that.

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Summary



Terminology covered

- Functional decomposition
- Timing budget
- Cooperative main loop
- Shared data protection
- Debugging checklist

Last chapter

Use this method in every design you build.